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PRODUCT CLASS

Fields:

```
private String productId;  
private String name;  
private double price;  
private int stockQuantity;  
private String description;
```

Methods:

```
public String getProductId()  
public String getName()  
public double getPrice()  
public int getStockQuantity()  
public void reducingStock(int quantity)  
public boolean isAvailable(int requiredQuantity)  
public void displayDetails()
```

ELECTRONICS(SUB CLASS FOR PRODUCT)

Extra Fields:

```
private int warrantyInMonths;  
private String brand;
```

private int powerRating;

Methods:

public int getWarranty()

public String getBrand()

@Override public void displayDetails()

Clothing (Subclass)

Extra Fields:

private String size;

private String fabric;

private String gender;

Methods:

public String getSize()

public String getFabric()

@Override public void displayDetails()

Grocery (Subclass)

Extra Fields:

private String expiryDate;

private boolean isOrganic;

private double weightInKg;

Methods:

public boolean isExpired()

@Override public void displayDetails()

User Class

Fields:

private String userId;

private String name;

private String email;

private String password;

private Address address;

Methods:

public void login()

public void logout()

public Cart getCart()

Cart Class

This is the brain of the cart system.

Fields:

```
private List<CartItem> items;
```

```
private double totalAmount;
```

Methods:

```
public void addProduct(Product product, int quantity)
```

```
public void removeProduct(String productId)
```

```
public void updateQuantity(String productId, int newQuantity)
```

```
public double calculateTotal()
```

```
public void clearCart()
```

```
public List<CartItem> getItems()
```

CartItem Class

It connects Product + Quantity.

Fields:

```
private Product product;
```

```
private int quantity;
```

Methods:

```
public double getSubtotal()
```

CartOperations (Interface)

This forces uniform behavior:

```
addProduct(Product p, int quantity);
```

```
removeProduct(String productId);
```

```
calculateTotal();
```

```
applyDiscount(Discount discount);
```

Cart implements this interface.

Discount Class

Fields:

```
private String discountId;
```

```
private double percentage;
```

Methods:

```
public double applyDiscount(double totalAmount)
```

Coupon Class (extends Discount)

Extra Fields:

```
private String couponCode;  
private double maxDiscountAmount;
```

Methods:

```
public boolean validateCoupon()  
@Override public double applyDiscount(double total)
```

Order Class

Fields:

```
private String orderId;  
private User user;  
private List<CartItem> orderedItems;  
private double finalAmount;  
private String status;
```

Methods:

```
public void generateInvoice()  
public void updateOrderStatus(String status)
```

Payment Class

Fields:

```
private String paymentId;  
private double amount;  
private String paymentMode;  
private String paymentStatus;
```

Methods:

```
public boolean processPayment()
```

Inventory Class

Fields:

```
private Map<String, Product> availableProducts;
```

Methods:

```
public void addProduct(Product product)  
public Product getProductById(String id)  
public void updateStock(String productId, int quantity)
```

Address Class

Fields:

```
private String street;  
private String city;
```

```
private String state;
```

```
private String pincode;
```